

SUHARSH KUMAR

Pune, Maharashtra

☎ +91-9661194055 ✉ suharshjha@gmail.com in Suharsh kumar 🌐 suharsh-kumar GFG: suharshjha

PROFILE SUMMARY

Computer Science undergraduate with strong expertise in Data Structures Algorithms and C++. Solved 600+ DSA problems using C++, with deep understanding of STL, OOP, and problem-solving techniques. Interested in building efficient and scalable systems using C++.

EDUCATION

Indian Institute of Information Technology, Pune	Aug 2023 – Present
Bachelor of Technology in Computer Science	CGPA: 8.26
Honors in Data Science	CGPA: 9.5
City Central High School (CBSE)	Samastipur, Bihar
Secondary School (2020): 93.4% Senior Secondary (2022): 92.4%	

SKILLS

Languages: C++ (Primary), C, Java, Python, SQL
C++ & Problem Solving: STL (vector, map, priority_queue), pointers, OOP in C++, recursion, DSA in C++
Core Computer Science: Data Structures & Algorithms, Operating Systems, DBMS
Backend (Basic): Spring Boot, REST APIs
Tools: Git, IntelliJ IDEA, VS Code
Soft Skills: Problem Solving, Analytical Thinking

PROJECTS

Smart Library Management System

Tech Stack: C++, OOPs, Java, Spring Boot, React.js, MySQL

- Designed a modular library system using object-oriented principles such as encapsulation and abstraction.
- Modeled core entities (books, users, transactions) for efficient data handling and fast retrieval.
- Optimized backend logic for operations like issuing, returns, and fine calculation with consistent performance.
- Developed scalable RESTful services ensuring efficient data flow and minimal latency.

LRU Cache System

Tech Stack: C++, STL

- Designed and implemented a caching system based on the Least Recently Used (LRU) eviction policy.
- Achieved $O(1)$ time complexity for get and put operations using a combination of Doubly Linked List and HashMap (unordered_map).
- Engineered efficient memory access and update operations through optimized pointer manipulation and data structure design.
- Implemented dynamic capacity handling with automatic eviction of least recently used elements to maintain performance.
- Extended functionality to track cache hits and misses for basic performance analysis.

Geospatial Nearest Neighbor Search System (C++)

Tech Stack: C++, STL

- Built a geospatial search system using the Great Circle Distance formula for accurate nearest neighbor computation.
- Implemented efficient nearest and top-k queries using priority_queue (heap) with $O(\log n)$ performance.
- Optimized computation using efficient data structures and memory handling with references/pointers to reduce overhead.
- Applied core Data Structures & Algorithms concepts to ensure scalability and performance.

ACHIEVEMENTS & CERTIFICATIONS

- Solved **600+** Data Structures & Algorithms problems using **C++ (STL)** on **GeeksforGeeks**, achieving a coding score of **2000+**.
- Maintained a **400+ day** coding streak on **GeeksforGeeks POTD**, demonstrating strong consistency in problem solving.
- **Finalist at TrustAI Ideathon 2026**, ranked among the top 5 teams nationally.
- Participated in **Adobe India Hackathon 2025** and **Google Solution Challenge 2025**.

ACTIVITIES & INTERESTS

- **Head, QuantNum – Mathematics Club, IIIT Pune (2025 – Present):** Leading quantitative problem-solving workshops, mathematical competitions, and analytical reasoning sessions.
- **Core Vocalist, Saaz – Music Club, IIIT Pune (2023 – Present):** Lead vocalist in college events; represented IIIT Pune in inter-college competitions.